

SUPERVISED CLASSIFICATION ALGORITHMS-KNN-NAÏVE BAYES

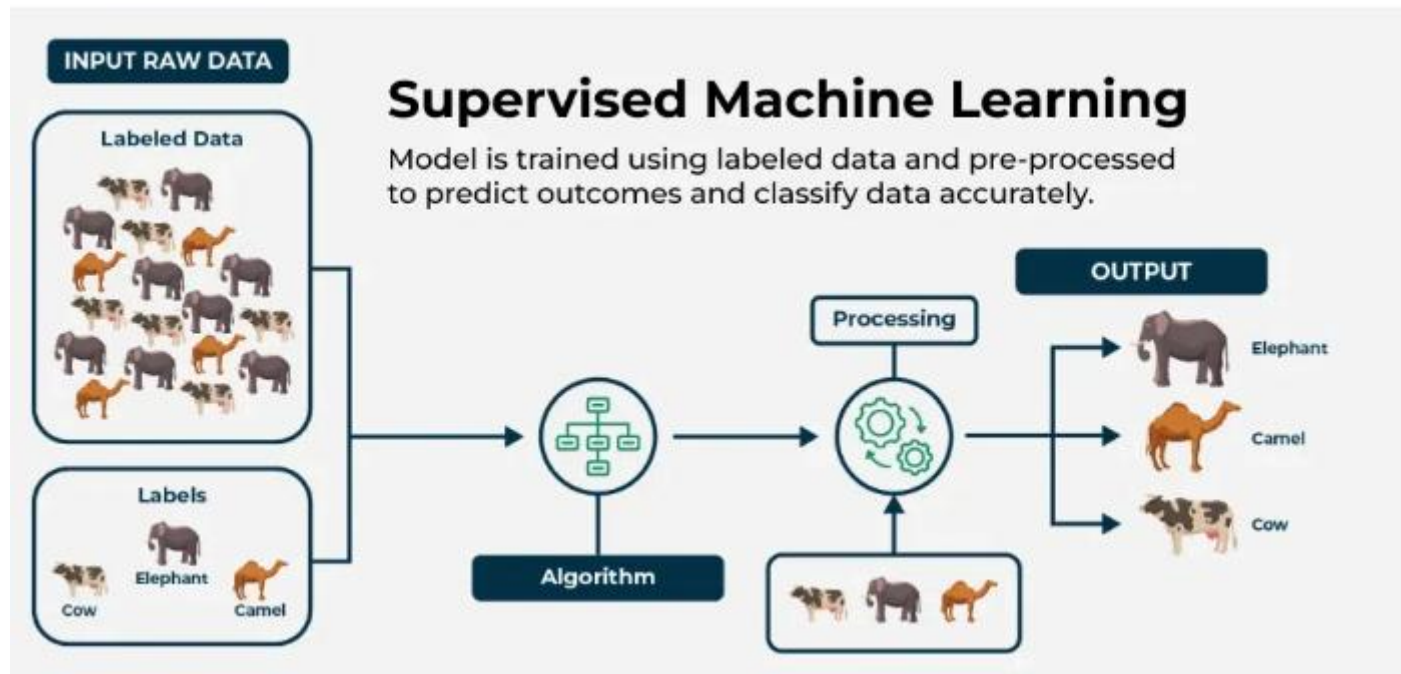
Week-7

Course Instructor : Mahrukh Khan



TYPES OF MACHINE LEARNING

- **Supervised learning** Algorithms learn from labelled data, where the input-output relationship is known.



K-NEAREST NEIGHBORS (K-NN)

- K-nearest neighbors (kNN) is a supervised machine learning algorithm that can be used to solve both classification and regression tasks.
- Classification by similarity
- kNN classifier determines the class of a data point by majority voting principle.



KNN-DISTANCE METRICS

- **Euclidean**

- **Straight-line distance between two points.**
- **Best for continuous features (height, weight, etc.).**

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- **Manhattan**

- **Distance if you can only move along grid lines (like streets in Manhattan).**
- **Better when features are independent or grid-like**

$$d = |x_1 - x_2| + |y_1 - y_2|$$

- **Cosine Similarity**

- **similarity measure (angle between vectors)**

$$\text{similarity} = \frac{A \cdot B}{||A|| ||B||}$$

$$A \cdot B = \text{dot product} = x_1 \cdot x_2 + y_1 \cdot y_2$$

$$||A|| = \text{magnitude of vector A} = \sqrt{x_1^2 + y_1^2}$$

$$||B|| = \text{magnitude of vector B} = \sqrt{x_2^2 + y_2^2}$$



ACTIVITY

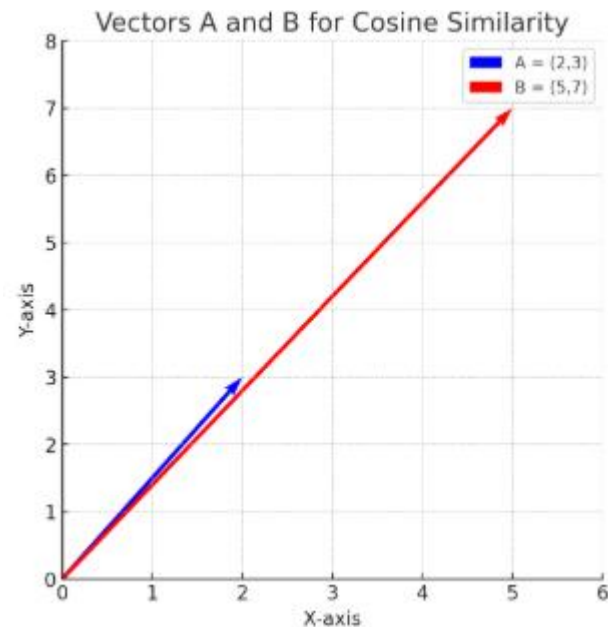
- Calculate for data points (2,3) and (5,7).
- $A \cdot B = (2)(5) + (3)(7) = 10 + 21 = 31$

$$\|A\| = \sqrt{2^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13}$$

$$\|B\| = \sqrt{5^2 + 7^2} = \sqrt{25 + 49} = \sqrt{74}$$

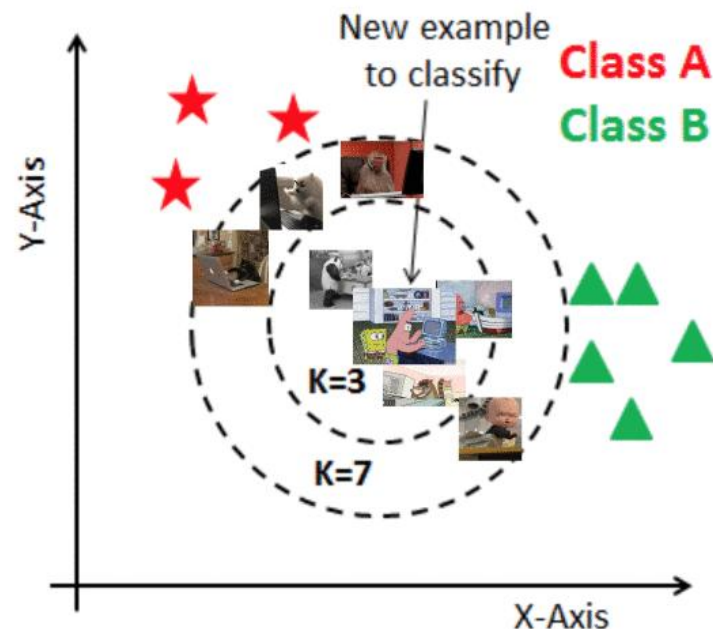
$$\cos \theta = \frac{31}{\sqrt{13} \cdot \sqrt{74}}$$

$$\cos \theta \approx \frac{31}{31.016} \approx 0.9995$$

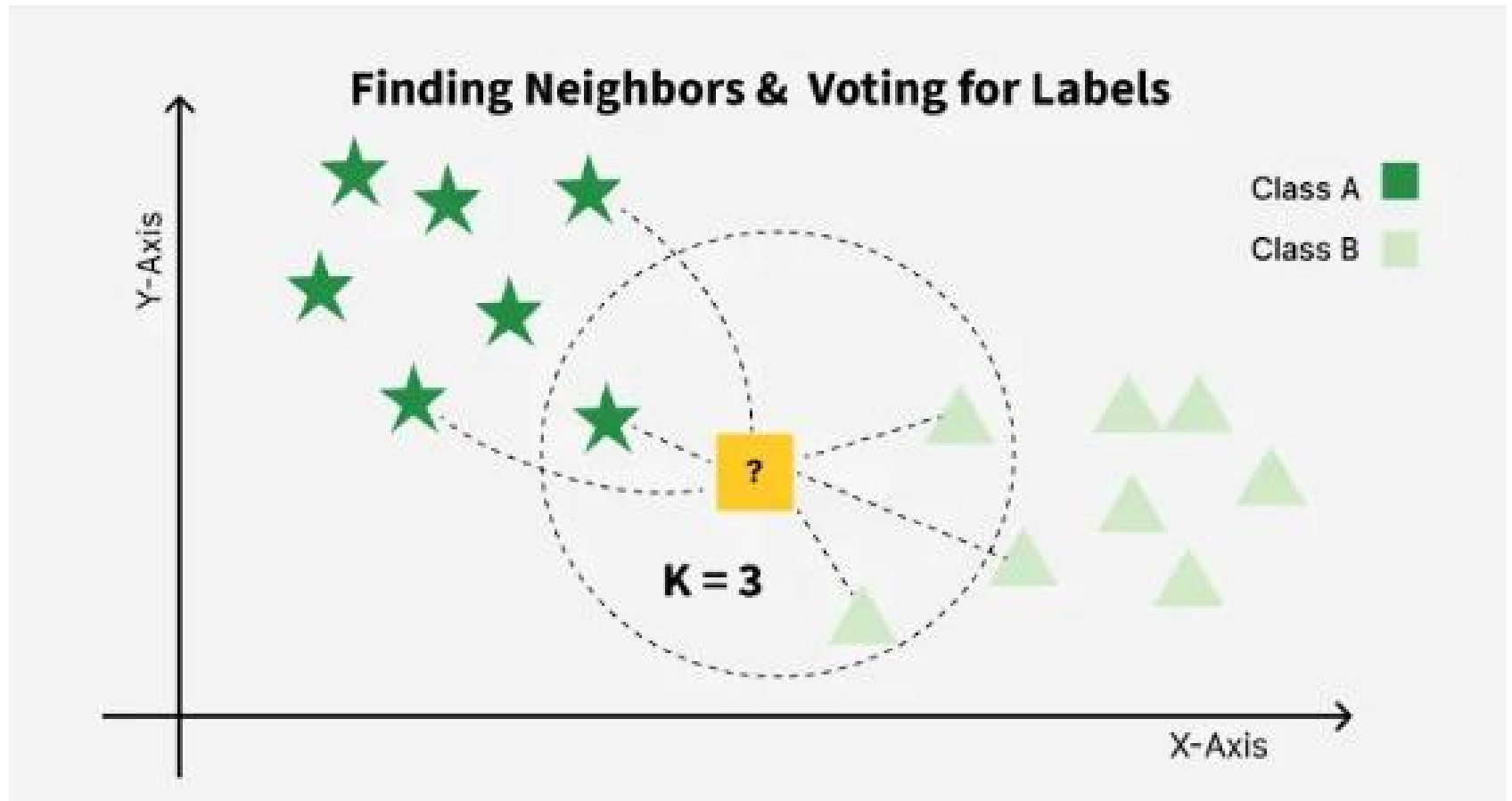


K-NEAREST NEIGHBORS (K-NN)

- It works by finding the "k" closest data points (neighbors) to a given input and makes a predictions based on the majority class (for classification) or the average value (for regression).
- k is just a number that tells the algorithm how many nearby points or neighbors to look at when it makes a decision.



K-NEAREST NEIGHBORS (K-NN)



KNN-EXAMPLE

- Below is a data that includes **age**, **gender** and the **class of sports** people play. Let's find in which class of people Angelina will lie whose k factor is 3 and age is 5.

NAME	AGE	GENDER	CLASS OF SPORTS
Ajay	32	0 (Male)	Football
Mark	40	0 (Male)	Neither
Sara	16	1 (Female)	Cricket
Zaira	34	1 (Female)	Cricket
Sachin	55	0 (Male)	Neither
Rahul	40	0 (Male)	Cricket
Pooja	20	1 (Female)	Neither
Smith	15	0 (Male)	Cricket
Laxmi	55	1 (Female)	Football
Michael	15	0 (Male)	Football



KNN-EXAMPLE

- K-Nearest Neighbors ($K = 3$): 3 nearest neighbors to Angelina are:
 - **Smith** (Cricket)- 10.5
 - **Michael** (Football)- 10.05
 - **Sara** (Cricket)- 11
- So according to KNN algorithm classifying based on Majority Vote, Angelina will be in the class of people who like cricket.



- **Large k** = biased toward majority.

KNN-HOW TO CHOOSE k ??

- **Basketball players:** tall and heavy
- **Football players:** average height and weight
- **Cricket players:** younger, lighter
 - We want to classify a **new student**.
- **Case 1: Very small k (e.g., $k = 1$)**
 - **Prediction only depends on 1 neighbour, if it happens to be an outlier, prediction will be wrong**
- **Case 2: Very large k (e.g., $k = 9$ for dataset of 10 points)**
- **Prediction covers almost the entire dataset. The minority class may get outvoted even if the new point is closer to them.**



KNN-HOW TO CHOOSE K??

- Larger K gives a smoother, more generalized boundary, but it may “wash out” small local clusters.
- With smaller K, the boundary is more jagged, meaning it follows local patterns closely (but can also be sensitive to noise).
- Common choice: $K = \sqrt{n}$ (n = number of samples).
- It should be large enough to avoid allowing noise and small enough not to be biased. This can be done by hyperparameter tuning with the help of Grid Search techniques.



KNN-ALGORITHM STEPS

1. Store all training data.
2. Choose value of K.
3. For a new data point:
 1. Calculate distance to all others.
 2. Pick K nearest neighbors.
4. For classification → majority vote.
5. For regression → average of neighbors.



KNN-ADVANTAGES

- Simple and easy to interpret
- Each of the distance functions has a different purpose based on the type of dataset. Based on the nature of features, it's possible to choose the best option -Manhattan and Euclidean for numeric, and Hamming for categorical features.
- KNN is a non-parametric algorithm and does not require any assumptions on the data distribution. This gives KNN an extra edge in specific settings where the data is highly unusual. This is the reason for KNN being the first choice when there is no prior knowledge or very little knowledge about the data distribution.
- Works well on classification with multiple classes.



KNN-DISADVANTAGES

- The efficiency of the algorithm declines very fast as the dataset grows, because the model needs to store all data points.
- Not memory efficient
- It suffers from skewed class distributions meaning if a specific class occurs frequently in the training set then it is most likely to dominate the majority voting of the new example.
- Sensitive to outliers. Outliers also have a vote!



APPLICATIONS OF KNN

- **Recommendation Systems:** Suggests items like movies or products by finding users with similar preferences.
- **Spam Detection:** Identifies spam emails by comparing new emails to known spam and non-spam examples.
- **Customer Segmentation:** Groups customers by comparing their shopping behaviour to others.
- **Speech Recognition:** Matches spoken words to known patterns to convert them into text.



TRAIN/TEST SPLIT

- If we train & test on the same data → model memorizes answers (like rote learning).
- If we test on new data → model has to generalize
- `train_test_split` divides the dataset into **two parts**:
 - **Training set (X_{train} , y_{train})**
→ Used to teach the model.
 - **Testing set (X_{test} , y_{test})**
→ Used to check how well the model learned.

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

- Split the dataset randomly, but do it in a way that's reproducible. Always give me the same result."



Guessing the Pet

Is it a cat or a dog in the box?

$P(\text{Dog}) = 0.5$



$P(\text{Cat}) = 0.5$



Initial Belief: 50/50 chances

- The probability we assign to an event **before** we see any new data or evidence.
- Think of it as your **initial belief**.

The updated probability of the event **after seeing new evidence**.

We have a CLUE



→ $(P(\text{Quiet} \mid \text{Cat})) = 80\% \text{ or } 0.8$

→ $(P(\text{Quiet} \mid \text{Dog})) = 30\% \text{ or } 0.3$



QUESTION: WHAT IS THE PROBABILITY THAT THE QUIET ANIMAL IS THE CAT?

- C : picked cat.
- D : picked dog.
- Q : animal is quiet.
- We want $P(C | Q)$.

$$P(C | Q) = \frac{P(Q | C) P(C)}{P(Q)}$$

$$P(Q) = P(Q | C)P(C) + P(Q | D)P(D)$$

- $P(C) = 0.5, P(D) = 0.5$.
- $P(Q | C) = 0.8$.
- $P(Q | D) = 0.3$.

$$P(Q) = 0.8 \times 0.5 + 0.3 \times 0.5 = 0.4 + 0.15 = 0.55.$$



BAYES THEOREM

- The probability of an event A based on the occurrence of another event B is termed Conditional Probability. It is denoted as $P(A|B)$ and represents the probability of A when event B has already happened.

$$P(C|Q) = \frac{0.8 \times 0.5}{0.55} = \frac{0.4}{0.55}.$$

- Prior $\rightarrow$ what you believed before data
- Posterior $\rightarrow$ what you believe after data

$$0.4 \div 0.55 = \frac{40}{55} = \frac{8}{11} \approx 0.727.$$

Applying Bayes Theorem to Find Probability



$$P(\text{Cat} | \text{Quiet}) = \frac{P(\text{Quiet Cat}) \times P(\text{Cat})}{P(\text{Quiet})} = 72.7\%$$



$$P(\text{Dog} | \text{Quiet}) = \frac{P(\text{Quiet Dog}) \times P(\text{Dog})}{P(\text{Quiet})} = 27.3\%$$

$$P(\text{cat} | \text{quiet}) \approx 72.7\%$$



EXAMPLE

- Box A has 2 red balls and 1 blue ball.
- Box B has 1 red ball and 2 blue balls.
- You pick a box at random (so each box has prior probability $1/2$), then draw one ball and it's red.
- Question: What is the probability the ball came from **Box A**?

$$P(A | R) = \frac{P(R | A) P(A)}{P(R)}$$

- where $P(R)$ = total probability of drawing a red ball from the random process.



$P(A) = 1/2$ (you chose Box A with probability 1/2).

$P(B) = 1/2$.

$P(R | A) = \frac{2}{3}$ (2 red out of 3 in A).

$P(R | B) = \frac{1}{3}$ (1 red out of 3 in B).

$$P(R) = P(R | A)P(A) + P(R | B)P(B) = \frac{2}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2}.$$

$$P(A | R) = \frac{\frac{2}{3} \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{1}{3} \times \frac{2}{1} = \frac{2}{3}.$$

$$P(\text{Box A} | \text{red}) = \frac{2}{3} \approx 66.7\%$$



BAYES THEOREM

- **Bayes' Theorem** is a mathematical formula used to determine the **conditional probability** of an event based on prior knowledge and new evidence.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

- $P(A | B)$ is the posterior probability, the likelihood of event A happening given that event B has already occurred.
- $P(B | A)$ is the likelihood, the probability of observing B if A is true.
- $P(A)$ is the prior probability, our initial belief about the chance of event A occurring.
- $P(B)$ is the total probability of event B happening (irrespective of A).



(BAYES' THEOREM — DISEASE TESTING EXAMPLE)

- A certain rare disease affects **1% of the population**. A medical laboratory has developed a test to detect this disease. The test is not perfect:
 - If a person **has the disease**, the test correctly shows **positive** with probability **90%** .
 - If a person **does not have the disease**, the test incorrectly shows **positive** with probability **5%** (false positive rate).
 - A person is randomly chosen from the population and their test result comes back **positive**.
- **Question:**
Using Bayes' Theorem, calculate the probability that this person actually has the disease. s



(BAYES' THEOREM — DISEASE TESTING EXAMPLE)

- **Given**

- (prior): 1% of the population have the disease. $P(Disease) = 0.01$
- (true positive rate): 90%. $P(Positive|Disease) = 0.90$
- (true negative rate): 95% → false positive rate = $1 - 0.95 = 0.05$.
 $P(Positive|No Disease) = 0.05$

- **Required**

- $P(Disease | Positive)$ = Probability that a person **actually has the disease** given that their test is positive.

- **Solution:**

- Here: $A = Disease$ and $B = Positive test$

- So, $P(Disease / Positive) = \frac{g(Positive / Disease) \cdot g(Disease)}{g(Positive)}$



(BAYES' THEOREM — DISEASE TESTING EXAMPLE)

- $P(\text{Positive}) = P(\text{Positive} \mid \text{Disease}) \cdot P(\text{Disease}) + P(\text{Positive} \mid \text{No Disease}) \cdot P(\text{No Disease})$
- $P(\text{Positive}) = (0.90)(0.01) + (0.05)(0.99)$
- $P(\text{Positive}) = 0.009 + 0.0495 = 0.0585$

$$P(\text{Disease} \mid \text{Positive}) = \frac{(0.90)(0.01)}{0.0585}$$

$$P(\text{Disease} \mid \text{Positive}) = \frac{0.009}{0.0585}$$

$$P(\text{Disease} \mid \text{Positive}) \approx 0.1538$$

•**Prior (before evidence):** 1% chance a random person has the disease.

•**Posterior (after positive test):** about **15.4% chance** the person actually has the disease.



FREQUENCY METHOD

- **Step 1: Count diseased and healthy people**

- Diseased = $1\% \times 100 = 1$ person.
- Healthy = $100 - 1 = 99$ people.

- **Step 2: Apply the test**

- **Among the 1 diseased person:**
- True positives = $0.90 \times 1 = 0.9$.
- False negatives = $0.1 \times 1 = 0.1$.

- **Step 3: Count total positive results**

- Total positives = True positives + False positives.
- $0.9 + 4.95 = 5.85$

- **Step 4: Compute posterior probability**

- $P(\text{Disease} / \text{Positive}) = \frac{0.9}{5.85}$
- Divide: $0.9 \div 5.85 \approx 0.1538 = 15.38\%$

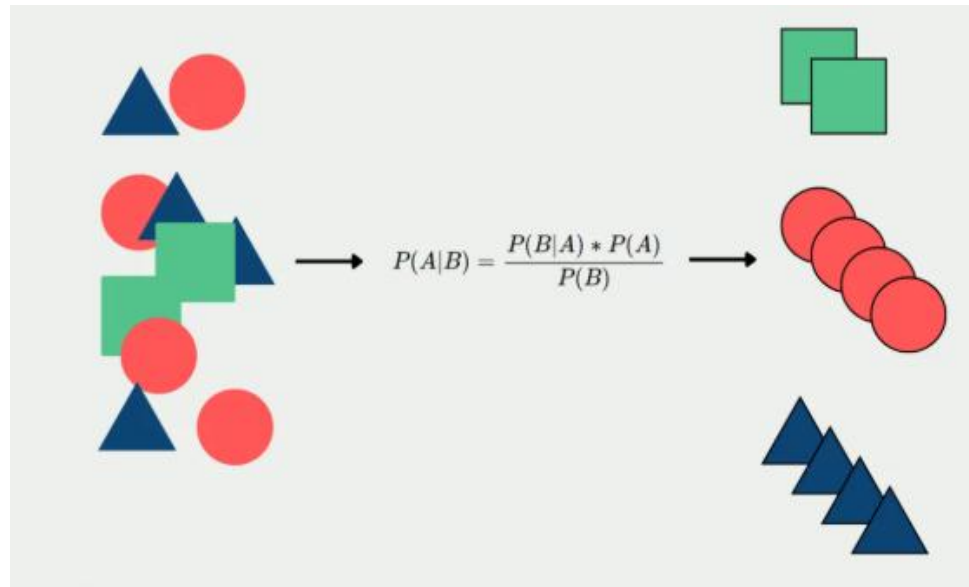
About **1 person is sick.**

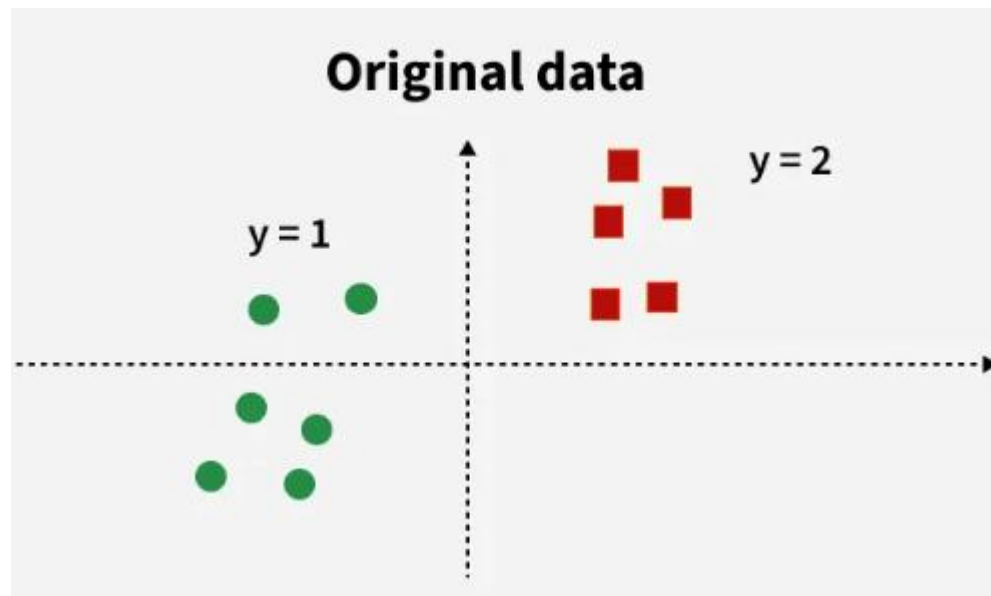
About **6 people test positive**, but only **1 of them is likely to be truly sick.**
So the chance a positive test means disease = **15.4%.**



NAIVE BAYES CLASSIFIER

- classification algorithm that predicts the category of a data point using probability.
- The main idea behind the Naive Bayes classifier is to use Bayes Theorem to classify data based on the probabilities of different classes given the features of the data.
- It is used mostly in high-dimensional text classification



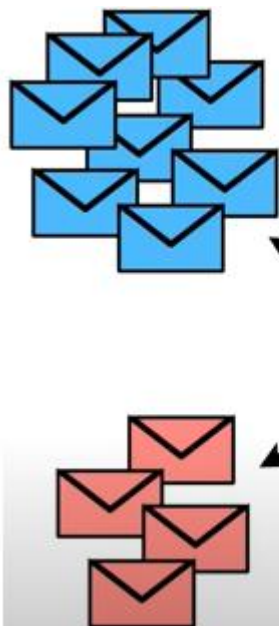




Imagine we received
normal messages from
friends and family...

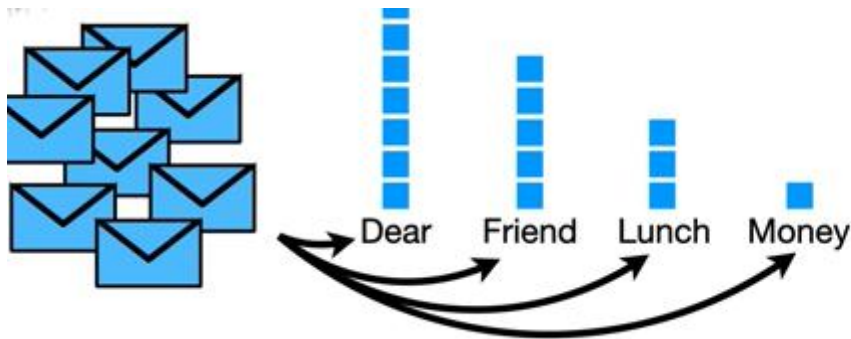


...and we also received
spam (unwanted
messages that are usually
scams or unsolicited
advertisements)...

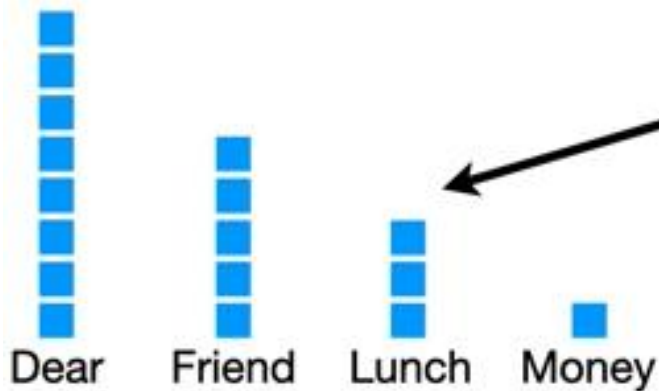


...and we wanted to filter
out the **spam** messages.



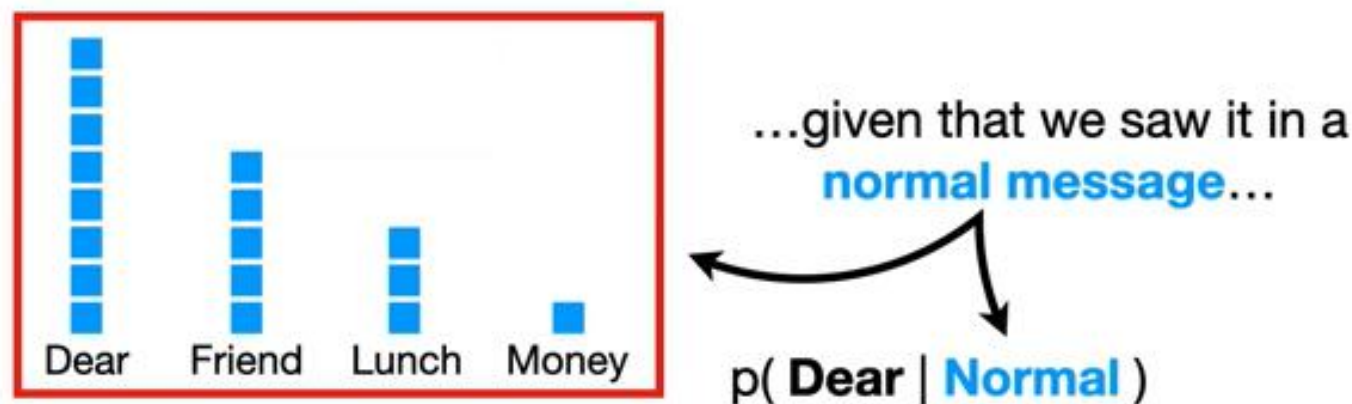
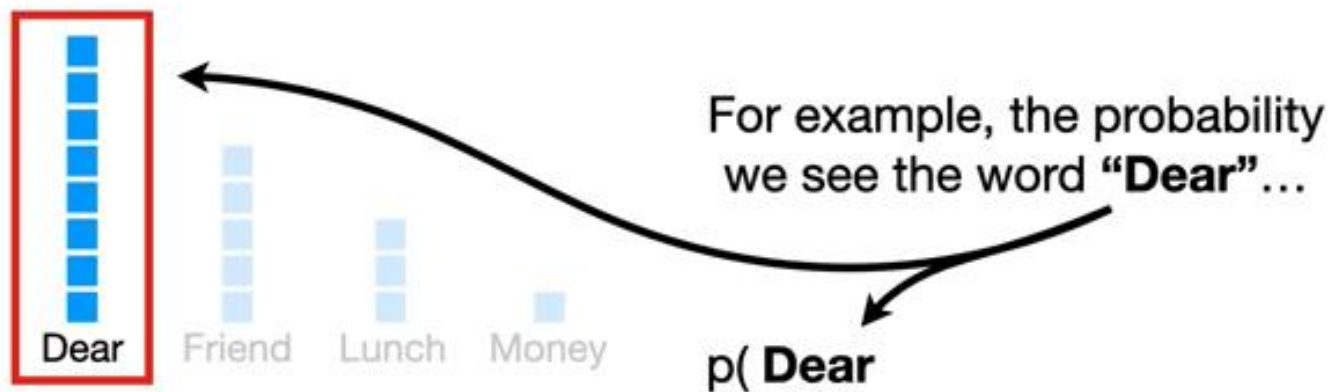


So, the first thing we do is make a **histogram** of all the words that occur in the **normal messages** from friends and family.



We can use the histogram to calculate the probabilities of seeing each word, given that it was in a **normal message**.





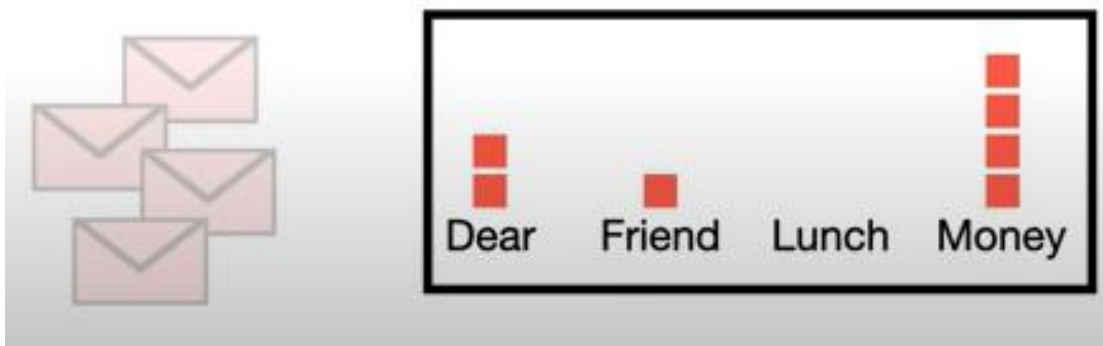
$$p(\mathbf{Dear} \mid \mathbf{Normal}) = \frac{8}{17}$$

$$p(\mathbf{Friend} \mid \mathbf{Normal}) = \frac{5}{17} = 0.29$$

$$p(\mathbf{Lunch} \mid \mathbf{Normal}) = \frac{3}{17} = 0.18$$

$$p(\mathbf{Money} \mid \mathbf{Normal}) = \frac{1}{17} = 0.06$$





$$\begin{aligned}p(\text{Dear} | \text{N}) &= 0.47 \\p(\text{Friend} | \text{N}) &= 0.29 \\p(\text{Lunch} | \text{N}) &= 0.18 \\p(\text{Money} | \text{N}) &= 0.06\end{aligned}$$

We have only calculated the probabilities of discrete, individual words, and not the probability of something continuous, like weight or height.



$$\begin{aligned}p(\text{Dear} | \text{S}) &= 0.29 \\p(\text{Friend} | \text{S}) &= 0.14 \\p(\text{Lunch} | \text{S}) &= 0.00 \\p(\text{Money} | \text{S}) &= 0.57\end{aligned}$$





$p(\text{Dear} | N) = 0.47$
 $p(\text{Friend} | N) = 0.29$
 $p(\text{Lunch} | N) = 0.18$
 $p(\text{Money} | N) = 0.06$

Dear Friend

?

?

Imagine we get a new message that said ..

And we want to decide if it's a normal message or a spam.



$p(\text{Dear} | S) = 0.29$
 $p(\text{Friend} | S) = 0.14$
 $p(\text{Lunch} | S) = 0.00$
 $p(\text{Money} | S) = 0.57$



$p(\text{Dear} | N) = 0.47$
 $p(\text{Friend} | N) = 0.29$
 $p(\text{Lunch} | N) = 0.18$
 $p(\text{Money} | N) = 0.06$

$$p(N) = \frac{8}{8 + 4} = 0.67$$



$p(\text{Dear} | S) = 0.29$
 $p(\text{Friend} | S) = 0.14$
 $p(\text{Lunch} | S) = 0.00$
 $p(\text{Money} | S) = 0.57$

Initial Guess->
Prior
Probability



Dear Friend

$$p(\text{Dear} | N) = 0.47$$

$$p(\text{Friend} | N) = 0.29$$

$$p(\text{Lunch} | N) = 0.18$$

$$p(\text{Money} | N) = 0.06$$

Now we multiply that initial guess by the probability that the word **Dear** occurs in a **normal message**...

$$p(N) \times p(\text{Dear} | N)$$

Dear **Friend**

...and the probability that the word **Friend** occurs in a **normal message**.

$$p(N) \times p(\text{Dear} | N) \times p(\text{Friend} | N)$$

$$0.67 \times 0.47 \times 0.29 = 0.09$$

However, technically, it is *proportional* to the probability that the message is **normal**, given that it says **Dear Friend**.

$$0.67 \times 0.47 \times 0.29 = 0.09 \propto p(N | \text{Dear Friend})$$



Now, just like we did before,
We start with an initial guess
about the probability that any
message, regardless of what
it says, is **spam**.

And since 4 of the 12
messages are **spam**,
our initial guess will be
0.33.

$$p(\mathbf{S}) = \frac{4}{4 + 8} = 0.33$$

$$p(\mathbf{S}) \times p(\mathbf{Dear} | \mathbf{S}) \times p(\mathbf{Friend} | \mathbf{S})$$

$$0.33 \times 0.29 \times 0.14 = 0.01$$

$$0.33 \times 0.29 \times 0.14 = 0.01 \propto p(\mathbf{S} | \mathbf{Dear Friend})$$

Let's try to classify this message ...

Lunch **Money Money Money Money**



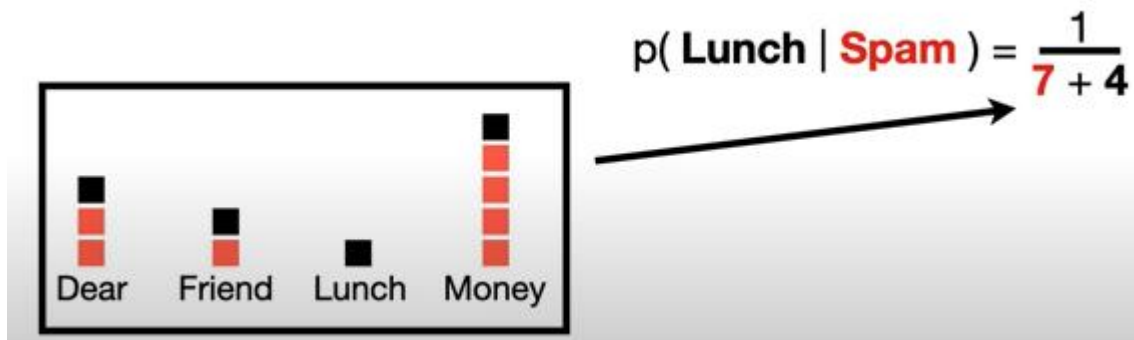
- We start with the initial probability that it's a normal message and calculate it,

$$p(\mathbf{N}) \times p(\mathbf{Lunch} | \mathbf{N}) \times p(\mathbf{Money} | \mathbf{N})^4 = 0.000002$$

- Now we calculate it, given the initial probability that it is spam,

$$p(\mathbf{S}) \times p(\mathbf{Lunch} | \mathbf{S}) \times p(\mathbf{Money} | \mathbf{S})^4 = 0$$

- Because the probability we see lunch in spam is 0, since it was not in training data. (in other words, if a message contains word lunch, it will not be classified as SPAM.)
- To work around this problem, we usually add 1 to the histograms we created earlier to each word, usually represented by a black box.



$$p(\mathbf{S}) \times p(\mathbf{Lunch} | \mathbf{S}) \times p(\mathbf{Money} | \mathbf{S})^4 = 0.00122$$

